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[123] CSE

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PART C: 16-MARK ESSAY & SYSTEM DESIGN QUESTIONS

Comprehensive Step-by-Step Architectural Breakdowns, Diagrams & Evaluated Solutions

Total Questions: 2 • Marks per Question: 16 Marks • Total Marks: 32 Marks

INDEX OF COVERED TOPICS & LEARNING UNITS:

1. Satellite Communication Transmission Fundamentals • 2. Satellite Link Budget and Transponder Performance

QUESTION 1 Topic: Satellite Communication Transmission Fundamentals **[16 Marks]**

Q1. Examine in detail the architectural role of uplinks, the calculation of Free Space Transmission Loss, and the definition of Output Back-Off (OBO) in High-Power Amplifiers within satellite communication systems.

Difficulty: HARD | Bloom: Analyze

[SOURCE GROUNDING]: 2_Mark_Questions_and_Answers (1) (Page 2) — "Q7. Identify the transmission specifications of the C-band transponder carried by a geostationary satellite fr..."

COMPREHENSIVE 16-MARK EVALUATED SOLUTION & TECHNICAL BREAKDOWN

1. Theoretical Introduction & Problem Motivation

Satellite communication systems rely on precise transmission links, path attenuation models, and amplifier linearity controls to ensure reliable signal delivery. The uplink serves as the foundational communication channel from the Earth station to the satellite, establishing the primary route for injected telemetry and traffic data. As signals propagate across vast distances through space, they experience geometric spreading known as Free Space Transmission Loss. Furthermore, onboard satellite transponders utilize High-Power Amplifiers (HPAs) like Traveling Wave Tube Amplifiers (TWTAs) which require Output Back-Off (OBO) to mitigate intermodulation distortion caused by non-linear saturation regions.

2. Architectural Principles & Step-by-Step Methodology

2.1 Uplink Purpose and Free Space Transmission Loss Mechanics

The uplink is designed to transmit carrier signals from earth stations to the spaceborne satellite payload. During propagation, electromagnetic waves spread spherically, causing power flux density to diminish inversely with the square of the distance. Free Space Transmission Loss quantifies this attenuation as a function of operating frequency and path length.

- Uplink establishes the initial frequency translation and routing path from ground to space.
- Free Space Loss depends strictly on frequency and propagation distance.
- Neglecting path attenuation leads to severe degradation in the overall carrier-to-noise ratio.

2.2 High-Power Amplifiers and Output Back-Off (OBO) Requirements

HPAs operate efficiently near saturation, but multiple simultaneous carriers cause intermodulation products if driven into non-linearity. Output Back-Off (OBO) is defined as the reduction in output power compared to the amplifier's saturation power level, ensuring linear operation and signal integrity.

- OBO prevents spectral regrowth and adjacent channel interference.
- Expressed in decibels relative to total saturation power.
- Essential for multi-carrier transponder configurations.

3. Architectural Schematic & Visual Model Representation

[Diagram] Satellite Transmission Link & HPA Non-Linearity Model

Schematic illustrating the uplink path from earth station, free space attenuation, and HPA back-off mitigation.

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[Earth Station Transmitter] --(Uplink)--> [Free Space Loss (FSL)] --> [HPA with OBO] --> [Satellite Transponder]
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4. Mathematical Formulations & Worked Numerical Proofs

Case 1: Calculation of Free Space Transmission Loss and OBO Adjustment

Step 1: Step 1: Identify frequency f = 6 GHz and distance d = 35,786 km for a geostationary link.
Step 2: Step 2: Apply the Free Space Loss formula L_fs = (4 * pi * d * f / c)^2.
Step 3: Step 3: Calculate OBO requirement given a saturation back-off requirement of 3 dB for linear multi-carrier operation.
Result: FSL approx 200 dB; Adjusted operating output power reduced by 3 dB OBO.

5. Critical Trade-offs, Advantages & Operational Invariants

[+] Advantage: Ensures reliable signal delivery from ground terminals to space payloads.
[+] Advantage: Protects transponder HPAs from destructive intermodulation distortion.
[-] Limitation: High free space loss requires large earth station antenna diameters and high transmitter power.
[-] Limitation: Output back-off reduces maximum attainable DC-to-RF conversion efficiency.

6. Synthesis, Conclusion & Practical Real-World Applications

Understanding uplinks, free space transmission loss, and output back-off is critical for designing robust, interference-free satellite communication links capable of high-fidelity signal transmission.

QUESTION 2 Topic: Satellite Link Budget and Transponder Performance **[16 Marks]**

Q2. Analyze the mathematical formulations and operational characteristics of Received Carrier Power (C), Combined Carrier-to-Noise (C/N) ratios in bent-pipe transponders, and satellite circuit interference.

Difficulty: HARD | Bloom: Analyze

[SOURCE GROUNDING]: 2_Mark_Questions_and_Answers (1) (Page 2) — "Q7. Identify the transmission specifications of the C-band transponder carried by a geostationary satellite fr..."

COMPREHENSIVE 16-MARK EVALUATED SOLUTION & TECHNICAL BREAKDOWN

1. Theoretical Introduction & Problem Motivation

Link budget analysis forms the backbone of satellite system engineering, dictating the physical dimensions, power levels, and modulation schemes required for reliable communication. Received carrier power (C) aggregates all transmitter, antenna, and path parameters into a unified power equation. In traditional bent-pipe transponders, the total carrier-to-noise ratio is synthesized from both uplink and downlink impairments. Additionally, utilizing multiple transponders within restricted frequency and spatial boundaries introduces complex circuit interference challenges that must be mitigated through frequency planning and polarization isolation.

2. Architectural Principles & Step-by-Step Methodology

2.1 Received Carrier Power (C) and Link Budget Parameters

The received carrier power in a satellite link budget incorporates Effective Isotropically Radiated Power (EIRP), free space path losses, atmospheric absorption, and receiving antenna gain. Each parameter directly scales the signal level available at the receiver input.

- Combines transmitter power and transmit antenna gain into EIRP.
- Accounts for path losses and polarization mismatch losses.
- Integrates receiving antenna gain to determine final available carrier power.

2.2 Bent-Pipe Transponder C/N Ratio and Circuit Interference

A bent-pipe transponder simply translates frequencies and amplifies signals without baseband demodulation. Consequently, noise from both the uplink and downlink segments combines reciprocally. Furthermore, multiple active transponders induce inter-modulation and cross-polarization interference.

- Total C/N is calculated using the reciprocal sum of uplink and downlink C/N ratios.
- Multi-transponder deployment causes inter-channel and co-channel interference.
- Requires robust filtering and guard bands to preserve signal quality.

3. Architectural Schematic & Visual Model Representation

[Diagram] Bent-Pipe Transponder Link & Noise Combination

Block diagram showing uplink noise addition, frequency translation, and downlink noise combination in a bent-pipe architecture.

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[Uplink + Noise_up] -> [Satellite Transponder (LNA/HPA)] -> [Downlink + Noise_down] -> [Total C/N Calculation]
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4. Mathematical Formulations & Worked Numerical Proofs

Case 1: Combined C/N Ratio Calculation for a Bent-Pipe Transponder

Step 1: Step 1: Determine uplink carrier-to-noise ratio (C/N)_up = 20 dB (linear ratio = 100).
Step 2: Step 2: Determine downlink carrier-to-noise ratio (C/N)_down = 13 dB (linear ratio = 19.95).
Step 3: Step 3: Calculate combined (C/N)_total using formula: 1 / (C/N)_total = 1 / (C/N)_up + 1 / (C/N)_down.
Result: Combined (C/N)_total evaluates to approximately 12.35 dB.

5. Critical Trade-offs, Advantages & Operational Invariants

[+] Advantage: Provides precise analytical framework for evaluating end-to-end link performance.

[+] Advantage: Bent-pipe architecture reduces satellite payload complexity and power consumption.
[-] Limitation: Accumulates noise from both uplink and downlink paths without regenerative cleanup.
[-] Limitation: Susceptible to inter-transponder interference when carrier density is high.

6. Synthesis, Conclusion & Practical Real-World Applications

Mastery of link budget parameters, combined C/N calculations, and interference mitigation ensures optimal satellite transponder utilization and high availability communication links.